

Module 3 Lesson 1

Quadratic Equations (6.5)

Objectives

In this lesson, we will learn how to

- multiply binomials using the FOIL method;
- factor trinomials;
- solve quadratic equations by factoring;
- solve quadratic equations using the quadratic formula;
- solve problems modeled by quadratic equations.

1. Multiplying Two Binomials Using the FOIL Method

A **binomial** is an algebraic expression containing two terms.

Examples of binomials include

$$x + 3, \quad x + 4, \quad 3x + 4, \quad 5x - 3.$$

FOIL Method

The FOIL method is used to multiply two binomials.

F: First terms

O: Outside terms

I: Inside terms

L: Last terms

Example 1: Multiply:

$$(x + 3)(x + 4).$$

Solution

Use FOIL:

$$(x + 3)(x + 4)$$

First terms:

$$x \cdot x = x^2.$$

Outside terms:

$$x \cdot 4 = 4x.$$

Inside terms:

$$3 \cdot x = 3x.$$

Last terms:

$$3 \cdot 4 = 12.$$

Therefore,

$$(x + 3)(x + 4) = x^2 + 4x + 3x + 12.$$

Combine like terms:

$$x^2 + 7x + 12.$$

2. Factoring Trinomials

A **trinomial** is a simplified algebraic expression that contains three terms.

For example,

$$x^2 + 7x + 12$$

is a trinomial.

Factoring means rewriting an expression as a product.

Factored Form and Trinomial Form

Using FOIL,

$$(x + 3)(x + 4) = x^2 + 4x + 3x + 12 = x^2 + 7x + 12.$$

So,

$$x^2 + 7x + 12$$

can be factored as

$$(x + 3)(x + 4).$$

3. Factoring $x^2 + bx + c$

For trinomials of the form

$$x^2 + bx + c,$$

we look for two numbers whose product is c and whose sum is b .

Strategy for Factoring $x^2 + bx + c$

To factor

$$x^2 + bx + c,$$

1. Write x as the first term of each binomial:

$$(x \quad)(x \quad).$$

2. List pairs of factors of the constant c .
3. Choose the pair whose sum is b .
4. Check by multiplying the factors using FOIL.

If no pair of integer factors works, the trinomial is called **prime** over the integers.

Example 2: Factor:

$$x^2 + 6x + 8.$$

Solution

We need two numbers whose product is 8 and whose sum is 6.
Factor pairs of 8:

$$1, 8$$

$$2, 4$$

Since

$$2 + 4 = 6,$$

we use 2 and 4.

Therefore,

$$x^2 + 6x + 8 = (x + 2)(x + 4).$$

Check:

$$(x + 2)(x + 4) = x^2 + 4x + 2x + 8 = x^2 + 6x + 8.$$

4. Factoring Trinomials with Leading Coefficient Greater Than 1

Some trinomials have a leading coefficient other than 1.

For example,

$$3x^2 - 20x + 28.$$

In this case, we must consider possible first terms and possible last terms.

General Factoring Idea

To factor

$$ax^2 + bx + c,$$

look for two binomials whose FOIL product gives the original trinomial.

Example 3: Factor:

$$3x^2 - 20x + 28.$$

Solution

The first terms must multiply to

$$3x^2.$$

So we can begin with

$$(3x \quad)(x \quad).$$

Since the constant term is positive and the middle term is negative, both signs should be negative.

Possible factor pairs of 28 include

$$1, 28; \quad 2, 14; \quad 4, 7.$$

Try

$$(3x - 14)(x - 2).$$

Use FOIL:

$$(3x - 14)(x - 2) = 3x^2 - 6x - 14x + 28.$$

Combine like terms:

$$3x^2 - 20x + 28.$$

Therefore,

$$3x^2 - 20x + 28 = (3x - 14)(x - 2).$$

5. Quadratic Equations

A **quadratic equation** in x is an equation that can be written in the form

$$ax^2 + bx + c = 0,$$

where a , b , and c are real numbers, and

$$a \neq 0.$$

Quadratic Equation

A quadratic equation has the form

$$ax^2 + bx + c = 0,$$

where $a \neq 0$.

6. Zero-Product Principle

The zero-product principle is very useful for solving quadratic equations by factoring.

Zero-Product Principle

If the product of two factors is zero, then at least one of the factors must be zero.

If

$$A \cdot B = 0,$$

then

$$A = 0$$

or

$$B = 0.$$

7. Solving Quadratic Equations by Factoring

Solving by Factoring

To solve a quadratic equation by factoring:

1. Rewrite the equation in the form

$$ax^2 + bx + c = 0.$$

2. Factor the quadratic expression.
3. Apply the zero-product principle.
4. Solve each resulting linear equation.
5. Check the solutions in the original equation.

Example 4: Solve:

$$x^2 - 2x = 35.$$

Solution

Move all terms to one side:

$$x^2 - 2x - 35 = 0.$$

Factor:

$$(x - 7)(x + 5) = 0.$$

Apply the zero-product principle:

$$x - 7 = 0$$

or

$$x + 5 = 0.$$

Solve:

$$x = 7$$

or

$$x = -5.$$

The solutions are

$$x = 7, \quad x = -5.$$

Check:

$$7^2 - 2(7) = 49 - 14 = 35.$$

$$(-5)^2 - 2(-5) = 25 + 10 = 35.$$

8. Quadratic Formula

Some quadratic equations are difficult or impossible to factor using integers.

In those cases, we can use the quadratic formula.

Quadratic Formula

The solutions of

$$ax^2 + bx + c = 0,$$

where $a \neq 0$, are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The expression under the square root,

$$b^2 - 4ac,$$

is called the discriminant.

Important Note

If

$$b^2 - 4ac < 0,$$

then the quadratic equation has no real solutions.

9. Solving with the Quadratic Formula

Example 5: Solve using the quadratic formula:

$$2x^2 + 9x - 5 = 0.$$

Solution

Identify

$$a = 2, \quad b = 9, \quad c = -5.$$

Use the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Substitute:

$$x = \frac{-9 \pm \sqrt{9^2 - 4(2)(-5)}}{2(2)}.$$

Simplify under the square root:

$$x = \frac{-9 \pm \sqrt{81 + 40}}{4}.$$

$$x = \frac{-9 \pm \sqrt{121}}{4}.$$

$$x = \frac{-9 \pm 11}{4}.$$

So,

$$x = \frac{-9 + 11}{4} = \frac{2}{4} = \frac{1}{2},$$

or

$$x = \frac{-9 - 11}{4} = \frac{-20}{4} = -5.$$

The solutions are

$$x = \frac{1}{2}, \quad x = -5.$$

10. Rewriting Before Using the Quadratic Formula

Sometimes the equation must first be rewritten in standard form.

Example 6: Solve using the quadratic formula:

$$2x^2 = 4x + 1.$$

Solution

Move all terms to one side:

$$2x^2 - 4x - 1 = 0.$$

Identify

$$a = 2, \quad b = -4, \quad c = -1.$$

Use the quadratic formula:

$$x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(2)(-1)}}{2(2)}.$$

Simplify:

$$x = \frac{4 \pm \sqrt{16 + 8}}{4}.$$

$$x = \frac{4 \pm \sqrt{24}}{4}.$$

$$x = \frac{4 \pm 2\sqrt{6}}{4}.$$

Reduce:

$$x = \frac{2 \pm \sqrt{6}}{2}.$$

The solutions are

$$x = \frac{2 + \sqrt{6}}{2}$$

and

$$x = \frac{2 - \sqrt{6}}{2}.$$

11. Application: Blood Pressure and Age

A person's normal systolic blood pressure, measured in millimeters of mercury, may depend on age.

The formula

$$P = 0.006A^2 - 0.02A + 120$$

models a male's normal systolic blood pressure, P , at age A .

Example 7: Find the age, to the nearest year, of a male whose systolic blood pressure is

125 mm Hg.

Solution

Substitute $P = 125$:

$$125 = 0.006A^2 - 0.02A + 120.$$

Move all terms to one side:

$$0.006A^2 - 0.02A - 5 = 0.$$

Identify

$$a = 0.006, \quad b = -0.02, \quad c = -5.$$

Use the quadratic formula:

$$A = \frac{-(-0.02) \pm \sqrt{(-0.02)^2 - 4(0.006)(-5)}}{2(0.006)}.$$

Simplify:

$$A = \frac{0.02 \pm \sqrt{0.0004 + 0.12}}{0.012}.$$

$$A = \frac{0.02 \pm \sqrt{0.1204}}{0.012}.$$

$$A \approx \frac{0.02 \pm 0.347}{0.012}.$$

The positive solution is approximately

$$A \approx 31.$$

Therefore, the male is approximately

$$31$$

years old.

12. More Practice

Example 8: Multiply:

$$(x - 5)(x + 2).$$

Solution

$$(x - 5)(x + 2) = x^2 + 2x - 5x - 10.$$

$$= x^2 - 3x - 10.$$

Example 9: Factor:

$$x^2 - 5x - 24.$$

Solution

We need two numbers whose product is -24 and whose sum is -5 .

The numbers are -8 and 3 .

Therefore,

$$x^2 - 5x - 24 = (x - 8)(x + 3).$$

Example 10: Solve by factoring:

$$x^2 + 3x - 10 = 0.$$

Solution

Factor:

$$x^2 + 3x - 10 = (x + 5)(x - 2).$$

So,

$$(x + 5)(x - 2) = 0.$$

Apply the zero-product principle:

$$x + 5 = 0$$

or

$$x - 2 = 0.$$

Thus,

$$x = -5$$

or

$$x = 2.$$

Summary

Key Ideas

- The FOIL method helps multiply two binomials.
- Factoring rewrites an expression as a product.
- A quadratic equation can be written as

$$ax^2 + bx + c = 0, \quad a \neq 0.$$

- The zero-product principle says that if $A \cdot B = 0$, then $A = 0$ or $B = 0$.
- Some quadratic equations can be solved by factoring.
- The quadratic formula is

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

- If $b^2 - 4ac < 0$, then the equation has no real solutions.
- Quadratic equations can be used to model real-world situations.